ECON 895 Coding Exercise 2

Spatial Techniques in Empirical Economics

Distributed Monday 26 October Due Monday 2 November, 7:20 PM

Two halves, ten points each. The build half asks you to make something. The audit half asks you to find what is wrong with something a machine made.

AI use. AI is permitted and expected on both halves. Any tool is allowed. Every submission includes a short *AI use note* saying what you asked the tools to do and what you checked before signing off. You sign the result and you own every number and every line.

Submit by email to `njohnsoL@gmu.edu` by 7:20 PM on Monday 2 November, from your Mason address, with the subject line "Exercise 2". Attach three files named with your last name: `lastname_exercise2.R` for the build half, `lastname_audit` in any readable format for the audit half, and `lastname_ai_note` for the AI use note.

Your build script must run in a fresh R session without errors. Load your packages at the top and read data from `data/` with paths relative to the repository root, as the weekly scripts do. It needs `gstat`, which `install.packages("gstat")` provides. I will run it.

Build half. Two surfaces for one country and lights for ten cities

Pick one country from the Black Death mortality data in `data/wk08/mort274.csv` that has at least ten recorded cities. France, Germany, Italy, Spain, and the United Kingdom qualify, and Belgium and Portugal sit at exactly ten. Steps 1–3 are for the cities of that country in `data/europe_cities_1801.csv`.

1. **Interpolate.** Using all 274 recorded cities as the source points, give every city in your country a mortality rate by inverse distance weighting. Choose the power by leave-one-out cross-validation over at least six values between 0.5 and 4, using the recorded cities of your country as the test points. Report the error at each power and the power you chose.

2. **Compare.** Give every city in your country the mortality of the nearest recorded city instead. Report how many cities differ from the inverse distance value by more than 20 points and name the three that differ most.

3. **Map.** One map of your country with the recorded cities as points and the interpolated cities colored by the inverse distance value.

4. **Lights.** Now switch continents. Sum the 2000 night lights in `data/wk08/africa_lights_2000.tif` within 25 km of each of the ten largest cities in `data/africa_cities.csv` and divide by the population within the same circle from `data/wk08/africa_popdens_2000.tif`. Report the ten values and say which circles contain cells at the sensor's ceiling.

5. **One paragraph.** Your interpolated map gives a mortality rate to every city in your country. For which cities is that number closest to a measurement and for which is it closest to a guess? Use the distance to the nearest recorded city and the leave-one-out error to answer.

Audit half. Five errors

The file `code/exercise2_audit.R` arrives with `git pull`. Run it from the repository root, as you run the weekly scripts. It reads like the code a language model writes from a one-sentence description of a task and five errors were planted in it.

The script is meant to interpolate Black Death mortality to all European cities with a cross-validated inverse distance power, report mortality by country, and compute night lights per person for every African country and the brightest of the 20 largest African cities.

It runs from start to finish in a fresh session without errors and the numbers it prints look plausible.

Some errors change the estimates and some change what the script reports. Find them. For each, write down

1. the line, or the lines,
2. what the code actually does, as against what it was meant to do,
3. why nothing complained,
4. the fix,
5. the check that would have caught it before it reached a table.

The fifth point is worth as much as the other four together.

Two notes. Running the script will not tell you what is wrong because none of the errors raises a message. Read the code. Every error in it is of a kind you have met this term, in a lecture, in a live demo, or in your own code.

The AI use note

Half a page at most. What you asked for, which tools, what came back, what you changed, and what you checked before you signed it. Include anything a model wrote that you did not trust and say why.

There is no penalty for using AI heavily and no credit for using it lightly. The grade is for the work and for the checking. The note itself carries no marks and it is required.

The exercise should take about one evening. The lights step is the slow one. Crop the raster to a box around each city before you sum.